

DRUG INTERACTIONS:

In elderly patients concurrently receiving diuretics, mainly thiazide, there appears to be an increased risk of thrombocytopenia with purpura. Occasionally, reports suggest that patients receiving pyrimethamine as malarial prophylaxis in doses exceeding 25 mg weekly, may develop megaloblastic anaemia should Co-trimoxazole be prescribed concurrently. Co-trimoxazole also has been shown to potentiate activity of strongly serum protein-bound drugs like oral hypoglycaemic or anticoagulant. It has been reported that sulphamethoxazole may displace the anticoagulant warfarin. Careful control of the anticoagulant therapy during treatment with Comazole is advisable. Comazole also may inhibit the hepatic metabolism of phenytoin. A 39% increase in phenytoin half-life and 27% decrease in the metabolic clearance rate of phenytoin has been observed following administration of Comazole of normal clinical dosages. If these two drugs are given concurrently it is important to remain alert for signs of phenytoin toxicity. Concurrent use of rifampicin and Comazole results in shortening of plasma half-life of trimethoprim.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION

Suspension

Children
6 – 12 years: 10 ml every 12 hours.
6 months to 5 years: 5 ml every 12 hours.
6 weeks to 5 months: 2.5 ml every 12 hours.

Tablet

It may be preferably to take this tablet with some food or drink to minimize the possibility of gastrointestinal disturbance.
Acute infection: Adult and children over 12 years old, standard dosage: 2 tablets every 12 hours.
Children 6 – 12 years, standard dosage: 1 tablet every 12 hour.
In acute infections, Comazole should be continued until the patient has been symptom-free for 2 days, the majority will require treatment for at least 5 days. If clinical improvement is not evident after 7 days therapy, the patient should be re-assessed.
For severe infection in all age groups, the dosage may be increased by 50%.
As an alternative to standard dosage for acute uncomplicated lower urinary tract infections, short-term therapy of 1 – 3 days duration has been shown to be effective.
The treatment is about 10-14 days. An identical daily dosage is used for 5 days in the treatment of shigellosis. Patients with impaired renal function need a reduced dosage.

Creatinine clearance ml /min	Recommended dosage
>30	Usual dosage
15 – 30	Half of the usual dosage
>15	Not recommended

SYMPTOMS AND TREATMENT FOR OVERDOSE AND ANTIDOTE(S):

Symptoms:
Nausea, vomiting, dizziness and confusion are likely signs/symptoms of overdosage. Bone marrow depression has been reported in acute trimethoprim overdosage.

Treatment:
If vomiting has not occurred, induction of vomiting may be desirable. Gastric lavage may be useful, though absorption from the gastrointestinal tract is normally very rapid and complete within approximately two hours. This may not be the case in gross overdosage. Dependent on the status of renal function administration of fluids is recommended if urine output is low.

Both trimethoprim and active sulfamethoxazole are moderately dialysable by hemodialysis. Peritoneal dialysis is not effective.

PACKING / PACK SIZES:

Suspension:HDPE bottles of 60ml,100 ml
Tablet: HDPE 20’s,100’s 500’s and 1000’s (for export only), Blister 100x10 Tablets

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:

For Suspension:
Store below 30°C, in a tight, light resistant container and protect from freezing.

For Tablet: Store at temperature below 30°C.
Store in a tight, light resistant container, protect from light

User instructions:

- Take this medicine on empty stomach, with full glass of water.
- Complete the treatment as prescribed by the physician.
- Keep all medicines out of reach of children.

SHELF LIFE: 3 years.

MANUFACTURER AND PRODUCT REGISTRATION HOLDER:

SM PHARMACEUTICALS SDN BHD (218620-M)
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Malaysia.

SM PHARMACEUTICALS SDN BHD

COMAZOLE

(SULPHAMETHOXAZOLE AND TRIMETHOPRIM TABLETS AND SUSPENSIONS)

DESCRIPTION:

Suspension: Reddish, raspberry flavoured, slightly viscous, uniform and free flowing, homogenous suspension.
Tablet: A white capsule shaped with deep concave surface and break-line, uncoated tablets.

COMPOSITION:

Suspension: Each 5 ml contains: Trimethoprim 40 mg, and Sulphamethoxazole 200 mg.
Tablet: Each tablet contains: Trimethoprim 80 mg, and Sulphamethoxazole 400 mg.

ACTIONS AND MODE OR MECHANISMS OF ACTION:

Sulphonamides are broad spectrum, bacteriostatic anti-infectives competitively inhibit a bacterial enzyme, dihydropteroate synthetase, that is responsible for incorporation of PABA into dehydrofolic acid, a cofactor for the synthesis of purines, thymidines and DNA. Trimethoprim is a bacteriostatic lipophilic weak base structurally related to pyrimethamine, which binds to and reversibly inhibits the bacterial enzyme dehydrofolate reductase, selectively blocking conversion of dehydrofolic acid. This depletes folate, an essential cofactor in the biosynthesis of nucleic acid and protein production. Bacterial dihydrofolate reductase is approximately 50 000 and 60 000 times more tightly bound by trimethoprim than the corresponding mammalian enzymes. Exerts its effect at a step in the folate biosynthesis immediately subsequently to the one in which sulphonamides exert their effect. When administered concurrently with sulphonamides, synergism occurs and is attributed to inhibition of tetrahydrofolate production at two sequential steps in its biosynthesis.

PHARMACOLOGY (SUMMARY OF PHARMACODYNAMICS AND PHARMACOKINETICS:

A combination which provides sequential and synergistic inhibition of bacterial folate synthesis. Sulphamethoxazole acts similarly to other sulphonamides while trimethoprim acts at a later step to inhibit the enzymatic reduction of dehydrofolic acid to tetrahydrofolic acid. The combination is active against many bacteria except anaerobes, *Pseudomonas aeruginosa* and many *S. faecalis*. It is also highly effective against protozoa and *Pneumocystis carinii*. Absorption: Orally, sulphamethoxazole and trimethoprim are 90 – 100% absorbed. Distribution: Sulphamethoxazole is widely distributed throughout the body tissues and fluids, including pleural, peritoneal, synovial and ocular fluid.

Trimethoprim is rapidly and widely distributed to various tissues and fluids, including kidneys, liver, spleen, bronchial secretion, saliva and prostatic tissues and fluids. It has also been demonstrated in bile, aqueous humor, bone marrow,

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and spongy but not compact bone, bowel mucosa, and seminal fluid. Both sulphamethoxazole and trimethoprim cross the placenta.
Protein binding: Sulphamethoxazole - approximately 66% in the plasma is protein bound.
Trimethoprim - approximately 50% in the plasma is protein bound.
Metabolism: Sulphamethoxazole - hepatic; primarily by acetylation to inactive metabolites which retain the toxicity of the parent compound. Some hepatic glucuronic conjugation may occur. Metabolism is increased with renal function impairment and decreased with hepatic failure.
Trimethoprim - hepatic; 10-20% metabolised to inactive metabolites.
Excretion: Sulphamethoxazole - renal by glomerular filtration, with some tubular secretion and re-absorption of both active drug and metabolites. Excretion is increased in alkaline urine. Small amounts are excreted in the faeces, breast milk, bile, and other body secretions.
Trimethoprim – renal; 40-60% excreted within 24 hours, of this amount 80-90% excreted unchanged and remainder excreted as inactive metabolites. Excretion increased in acid urine and decreased in alkaline urine.
Small amounts excreted in the faeces (approximately 4%), bile, and breast milk.

TOXICOLOGY:

Cross sensitivity:
Patients intolerant of one sulphonamide may be intolerant of other sulphonamides also. Patient intolerant of frusemide, thiazide diuretics, sulphonylureas or carbonic anhydrase inhibitors may be intolerant of sulphonamides also.

Pregnancy / reproduction:
Sulphonamides and trimethoprim cross the placenta. Although sulphonamides may displace bilirubin from protein binding sites in the foetal plasma, significant hyperbilirubinaemia does not usually occur in the neonate because of maternal hepatic conjugation of bilirubin.

Breast-feeding:
Sulphonamides and trimethoprim are excreted in breast milk. However approximately 0.2% of the maternal dose of trimethoprim appears in breast milk. Although sulphonamides may displace bilirubin from protein binding sites of the foetal plasma, hyperbilirubinaemia does not usually occur except as a remote possibility during the first two week postpartum. Sulphonamides may cause haemolytic anaemia in glucose-6-phosphate dehydrogenase (G6PD) deficient neonates.

INDICATIONS:

For the treatment of Urinary tract infections (Upper and lower), Respiratory infections, Genital tract infections, Gastrointestinal tract infections, Skin and wound infections, Other bacterial infections caused by sensitive organisms, Fungal skin infections; South American blastomycosis.

CONTRAINDICATIONS:

Comazole is contraindicated in patients with marked liver parenchymal damage. It is also contraindicated in patients with severe renal insufficiency when repeated determinations of the plasma concentration cannot be made.
Except in rare circumstances Comazole should not be given to patients with serious haematological disorders. The combination has occasionally been administered to patients receiving cytotoxic agents for the treatment of leukaemia,

without evidence of any adverse effect on the bone marrow or peripheral blood.
Comazole should not be administered to patients with a history of hypersensitivity to sulphonamides or trimethoprim.
For safety reasons Comazole is contraindicated during pregnancy. If pregnancy cannot be excluded, possible risks should be balanced against the expected therapeutic effect.
Comazole should not be given to premature and newborn infants during the first week of life.
Comazole is contraindicated in G6PD (Glucose-6-phosphate dehydrogenase deficiency). Treatment must be immediately discontinued on the appearance of a skin rash or any element of blood count reduced.

SIDE EFFECTS / ADVERSE REACTIONS:

At the recommended dosage, Comazole is usually well tolerated. Of the reported adverse effects, most are mild and comprised nausea (with or without vomiting) and skin rashes. As with great variety of other drugs, Comazole has isolated cases been associated with the Steven –Johnson syndrome.
Haematological changes have been reported, the majority being mild, asymptomatic and reversible on withdrawal of the drug. The changes mainly took the form of leukopenia, neutropenia and thrombocytopenia. Very rarely, agranulocytosis, megaloblastic anaemia and purpura occurred. Although trimethoprim and sulphamethoxazole are excreted in breast milk, administration of this combination to nursing mothers represents a negligible risk to the infants.
The most common side effects are gastrointestinal disturbances (nausea, vomiting, anorexia) and also skin rashes. Also includes glossitis, pancreatitis, stomatitis, diarrhoea and rarely cases of pseudomonas enterocolitis.

Vascular disorders
Frequency ‘not known’: circulatory shock

PRECAUTIONS / WARNINGS:

In patients with impaired renal function, the dosage should be reduced or the interval between doses prolonged in order to prevent cumulation in the blood. Determination of plasma drug concentrations is recommended in such patients.
Regular blood count is advisable whenever Comazole is given for prolonged periods, especially in elderly, where there is possibility of haematological changes indicative of folic acid deficiency; these are reversible by folinic acid therapy.
An adequate output should be maintained at all times. Treatment must be discontinued immediately if skin rash appears.
Blood dyscrasias; death associated with the administration of sulphonamides have been reported from hypersensitivity reactions, agranulocytosis, aplastic anaemia and other blood dyscrasias.
Clinical signs such as sore throat, fever, pallor, purpura or jaundice may be early indications of blood disorders.

Complete blood count should be done frequently in patients receiving sulphonamides. Trimethoprim has been reported to interfere with haematopoiesis in occasional patients.

Group A-beta haemolysis streptococcal tonsillopharyngitis; Patients treated with trimethoprim and sulphamethoxazole have a greater incidence of bacteriological failure than those treated with penicillin, as evidenced by failure to eradicate this organism from the

tonsillopharyngeal area.
Superinfections: Possible overgrowth of non-susceptible organisms, including fungi, should be looked for when any sulphonamide is administered. Should this occur treatment should be discontinued and appropriate therapy instituted.
Severe allergy or bronchial asthma: Sulphonamides should be given with caution to these patients, with a critical appraisal of benefit versus risk.
Comazole also should be given with caution to patients with impaired hepatic function and to those patients with severe allergy or bronchial asthma. In G6PD, hemolysis may occur (dose related).

Because of the risk of Crystalluria , an adequate fluid intake should be maintained and the administration of alkalis may be necessary, if very large doses are used.

Fatalities associated with administration of sulphonamides and trimethoprim, either alone or combination, have occurred due to severe reactions, including Steven-Johnson Syndrome, toxic epidermal necrolysis and other reactions. This drug should be discontinued at the first appearance of skin rash or any sign of adverse reaction.

Respiratory toxicity Very rare, severe cases of respiratory toxicity, sometimes progressing to Acute Respiratory Distress Syndrome (ARDS), have been reported during co-trimoxazole treatment. The onset of pulmonary signs such as cough, fever and dyspnoea in association with radiological signs of pulmonary infiltrates, and deterioration in pulmonary function may be preliminary signs of ARDS. In such circumstances, co-trimoxazole should be discontinued and appropriate treatment given.

Circulatory shock
Cases of circulatory shock, often accompanied by fever and not responding to standard treatment for hypersensitivity, have been reported with sulfamethoxazole + trimethoprim, mainly in immunocompromised patients.

Use in Pregnancy and lactation:

The safety of Comazole in human pregnancy has not been established. At doses greatly in excess of the recommended human therapeutic dose, trimethoprim has been reported to be teratogenic in rats with effects typical of a folate antagonist and preventable by administration of dietary folate. No significant drug-related malformations have been demonstrated in rabbits, but at doses approximately 10 times in excess of human therapeutic dose, an increase in foetal death was noted. Despite the excretion of trimethoprim and sulphamethoxazole into breast milk, the administration of Comazole to lactating women represents a negligible risk to the suckling infants.
Comazole should not be given during pregnancy especially in late pregnancy because of risk of kernicterus. It also should be used with caution in nursing mothers because it can pass the placenta and is excreted in milk and may cause kernicterus.